Redox & Electron Transfer

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Written by: Alexandra Brennan
Reviewed by: Stewart Hird
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Redox & electron transfer

Extended tier only

- Redox reactions can also be defined in terms of electron transfer
- **Oxidation** is a reaction in which an element, ion or compound **loses electrons**
  - The **oxidation number** of the element is **increased**
  - This can be shown in a half-equation, e.g. when silver reacts with chlorine, silver is **oxidised** to silver ions:

$$Ag \rightarrow Ag^{+} + e^{-}$$

- **Reduction** is a reaction in which an element, ion or compound **gains electrons**
  - The **oxidation number** of the element is **decreased**
  - This can be shown in a half-equation, e.g. when oxygen reacts with magnesium, oxygen is **reduced** to oxide ions:

$$O_2 + 4e^{-} \rightarrow 2O^{2-}$$

- For example, when iron reacts with a compound of copper such as copper sulfate, a displacement reaction occurs

iron + copper sulfate → iron(II) sulfate + copper

$$\mathbf{Fe + CuSO_4 \rightarrow FeSO_4 + Cu}$$

- We can write this as an ionic equation

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Ask questions and get help with exactly what you're revising.
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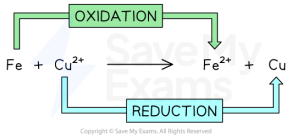
change

$$Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$$

- The iron atom has lost electrons to become a positive ion, so has been oxidised
- The positive copper ion has gained electrons to become an atom, so have been reduced

The redox reaction between Fe and Cu^{2+}



The Fe atom is oxidised (loses electrons) and the Cu^{2+} ion is reduced (gains electrons)

Worked Example

Which change in the following equation is oxidation?

$$V^{3+} + Fe^{3+} \rightarrow V^{4+} + Fe^{2+}$$

Answer:

- **Step 1** - Identify the changes for each species:
  - V^{3+} to V^{4+}
    - V^{3+} has lost 1 electron
  - Fe^{3+} to Fe^{2+}
    - Fe^{3+} has gained 1 electron
- **Step 2** - Identify each change as either oxidation and reduction
  - V^{3+} to V^{4+} is oxidation
  - Fe^{3+} to Fe^{2+} is reduction
- Therefore, V^{3+} has been oxidised

Examiner Tips and Tricks

Use the mnemonic **OIL-RIG** to remember oxidation and reduction in terms of the movement of electrons:

- **O**xidation **I**s **L**oss
- **R**eduction **I**s **G**ain.

Identifying redox reactions

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- The **oxidation number** is a number assigned to an atom or ion in a compound
- It shows the number of electrons that an atom has lost, gained or shared in forming a compound
    - So, the oxidation number helps you to keep track of the movement of electrons in a redox process

- It is written as a +/- sign followed by a number
    - Positive oxidation number = loss of electrons
    - Negative oxidation number = gain of electrons
    - For example, aluminium in a compound usually has the oxidation number of +3 indicating it has lost 3 electrons
    - **Careful:** It is easy to confuse oxidation number with charge which is written by a number followed by a +/- sign)

- A few simple rules help guide you through the process of determining the oxidation number of any element

Rules for assigning oxidation numbers

	Rule	Example
1	The oxidation number of any uncombined element is zero	H_2 Zn O_2
2	Many atoms or ions have fixed oxidation numbers in compounds	Group 1 elements are always +1 Group 2 elements are always +2 Fluorine is always −1 Hydrogen is +1, except in hydrides like NaH where it is −1 Oxygen is −2, except in peroxides where it is in −1 and in F_2O where it is +2
3	The oxidation number of an element in a monoatomic ion is always the same as the charge	Zn^{2+} = +2 Fe^{3+} = +3 Cl^- = −1
4	The sum of the oxidation numbers in a compound is zero	NaCl Na = +1 Cl = −1 Sum of oxidation numbers = 1 − 1 = 0
5	The sum of the oxidation numbers in an ion is equal to the charge on the ion	SO_4^{2-} S = +6 Four O atoms = 4 x (−2) = −8 Sum of oxidation numbers = 6 − 8 = −2

	oxidation number	O = +2

- Redox reactions can be identified by the changes in the **oxidation number** when a reactant goes to a product

Worked Example

The equation for the reaction between chlorine and potassium iodide is shown below.

$$Cl_2 + 2KI \rightarrow 2KCl + I_2$$

Identify which species has been:

1. Oxidised
2. Reduced

Answer:

1. The species that has been oxidised is iodine
   - $2I^- \rightarrow I_2 + 2e^-$
   - The oxidation number of I^- is -1
   - The oxidation number of iodine in I_2 is 0
   - The oxidation number has increased so the iodide ions have been oxidised / lost electrons
2. The species that has been reduced is chloride ions
   - $Cl_2 + 2e^- \rightarrow 2Cl^-$
   - The oxidation number of chlorine as Cl_2 is 0
   - The oxidation number of Cl^- is -1
   - The oxidation number has decreased so the Cl_2 has been reduced / gained electrons

Identifying redox reactions by colour changes

- The tests for redox reactions involve the observation of a colour change in the solution being analysed
- Two common examples are acidified potassium manganate(VII), and potassium iodide
- Potassium manganate(VII), $KMnO_4$, is an **oxidising agent** which is often used to test for the presence of **reducing** agents
- When acidified potassium manganate(VII) is added to a reducing agent its colour changes from **purple** to **colourless**

Diagram to show the colour change when potassium manganate(VII) is added to a reducing agent

- Potassium iodide, KI, is a **reducing** agent which is often used to test for the presence of **oxidising** agents
- When added to an acidified solution of an oxidising agent such as aqueous chlorine or hydrogen peroxide (H_2O_2), the solution turns a **red-brown** colour due to the formation of iodine, I_2

Diagram to show the colour change when potassium iodide is added to an oxidising agent

- The potassium iodide is oxidised as it loses electrons
- The hydrogen peroxide is reduced
- Therefore, potassium iodide is acting as a **reducing agent**

Oxidising & reducing agents

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What is an oxidising agent?

- An oxidising agent is a substance that **oxidises** another substance, and becomes **reduced** in the process
- An oxidising agent **gains** electrons as another substance loses electrons
- Common examples include hydrogen peroxide, fluorine and chlorine

What is a reducing agent?

gains electrons
- Common examples include carbon and hydrogen
- The process of reduction is very important in the chemical industry as a means of extracting metals from their ores

Identifying oxidising and reducing agents

$$CuO + H_2 \rightarrow Cu + H_2O$$

- Hydrogen is reducing the CuO
- Hydrogen is itself oxidised as it has gained oxygen / lost electrons
- So, the **reducing agent** is **hydrogen**:

$$H_2 \rightarrow 2H^+ + 2e^-$$

- CuO is reduced by hydrogen
  - This means that the hydrogen is oxidised by CuO
- CuO is reduced as it has lost oxygen / gained electrons
- So, the **oxidising agent** is **copper oxide**

$$Cu^{2+} + 2e^- \rightarrow Cu$$

Worked Example

When iron reacts with bromine to form iron(II) bromide, a redox reaction reaction occurs:

$$Fe + Br_2 \rightarrow FeBr_2$$

Which species is acting as the reducing agent in this reaction?

Answer

1. **Step 1** - Write half equations to work out what has gained/lost electrons
   - $Fe \rightarrow Fe^{2+} + 2e^-$
   - $Br_2 + 2e^- \rightarrow 2Br^-$
   - Fe loses electrons; Br_2 gains electrons
2. **Step 2** - Deduce what has been oxidised/reduced (remember OIL RIG)
   - Fe has been oxidised as it has lost electrons
   - Br_2 has been reduced as it has gained electrons
3. **Step 3** - Identify the reducing agent
   - Fe is the reducing agent as it has been oxidised by losing electrons and caused Br_2 to be reduced as it gained electrons

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